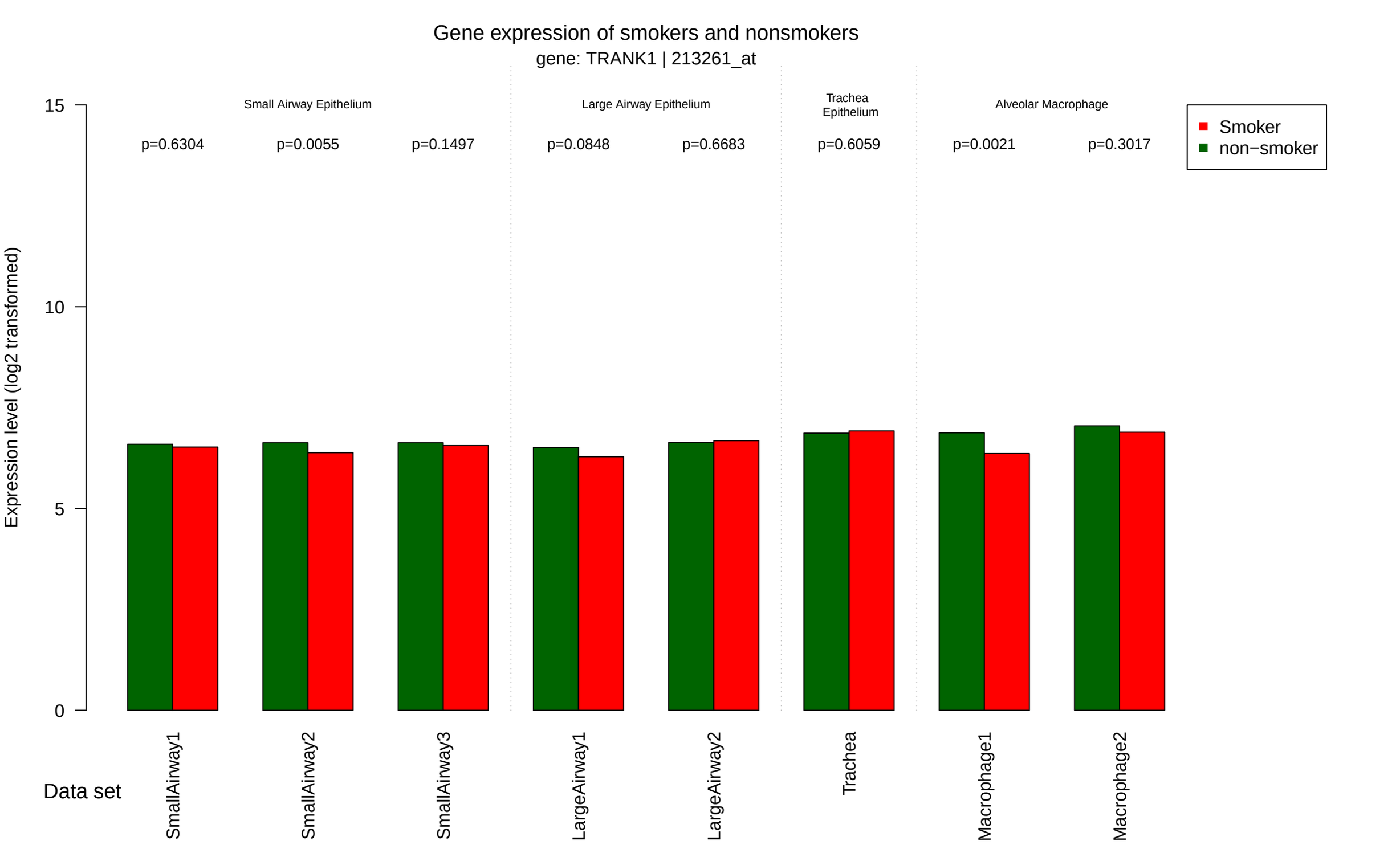
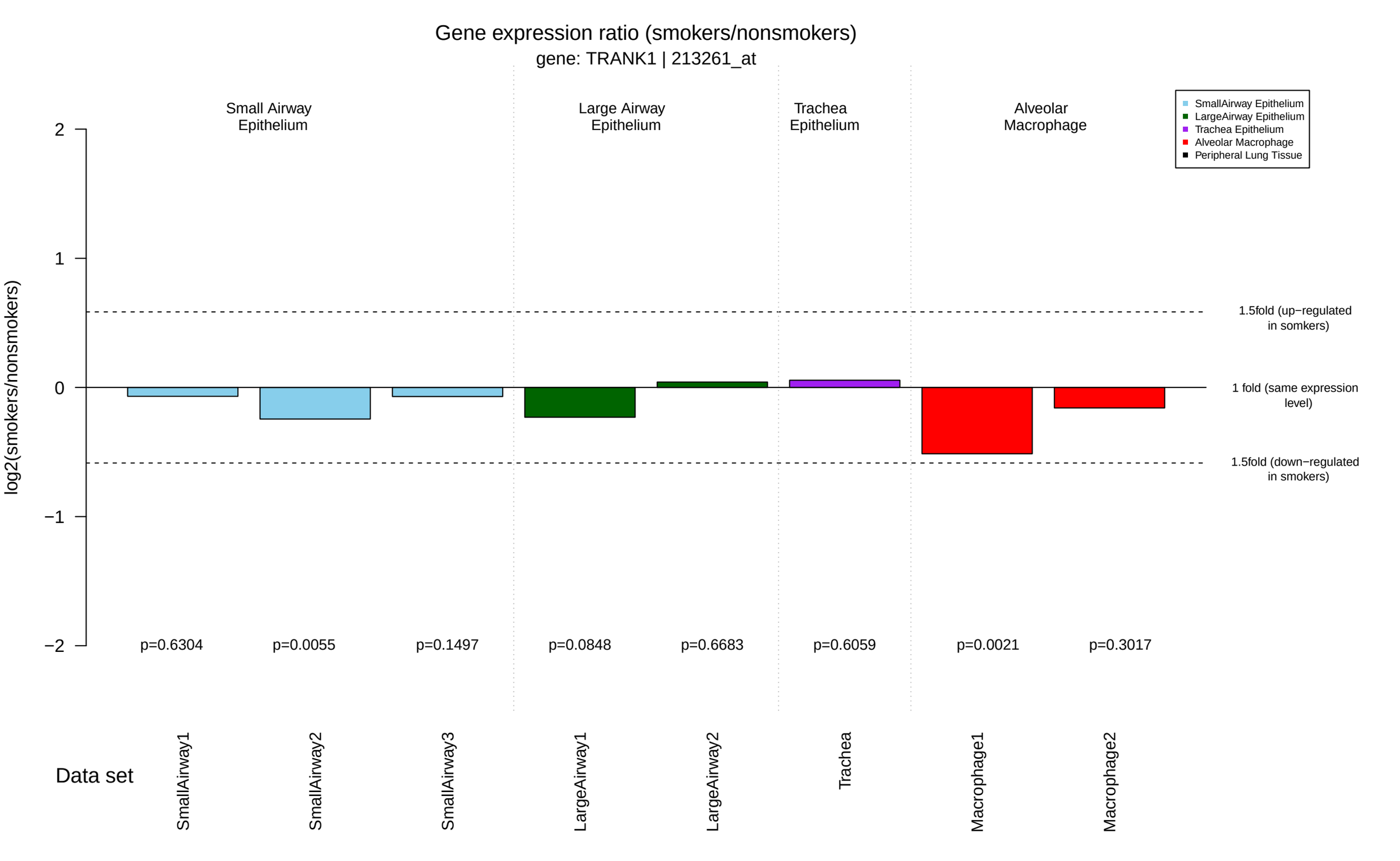
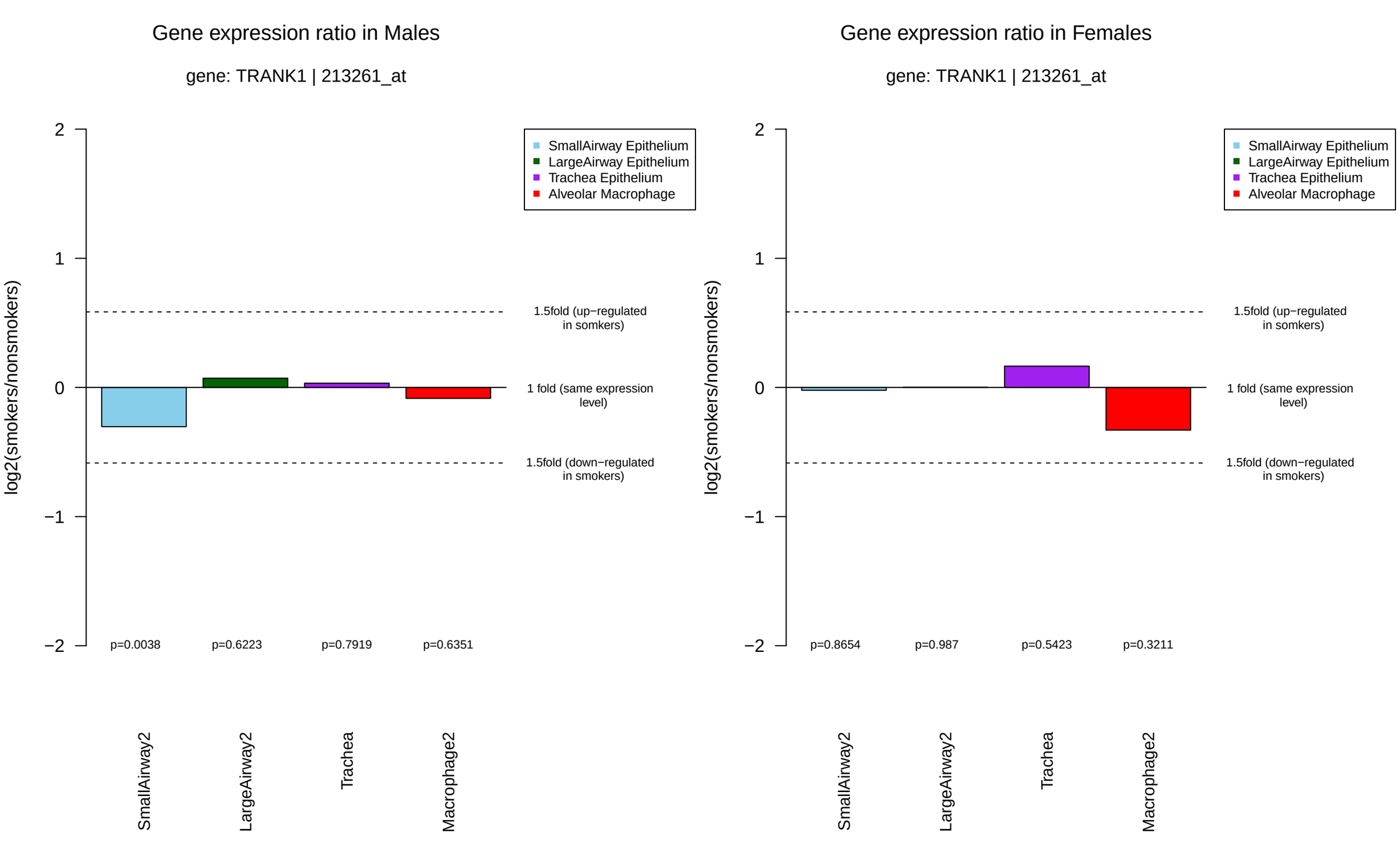
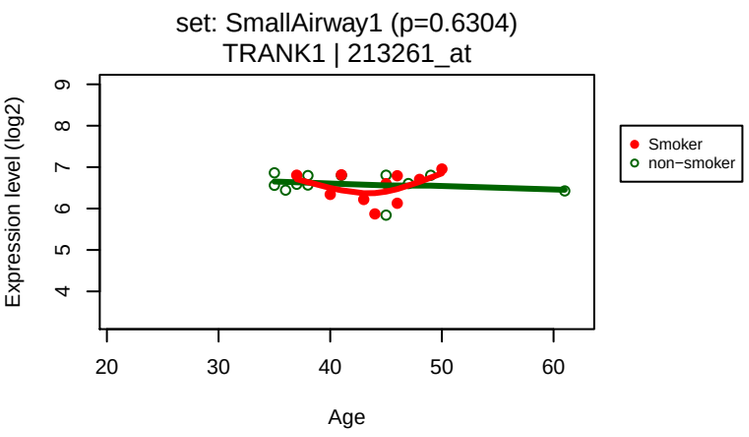
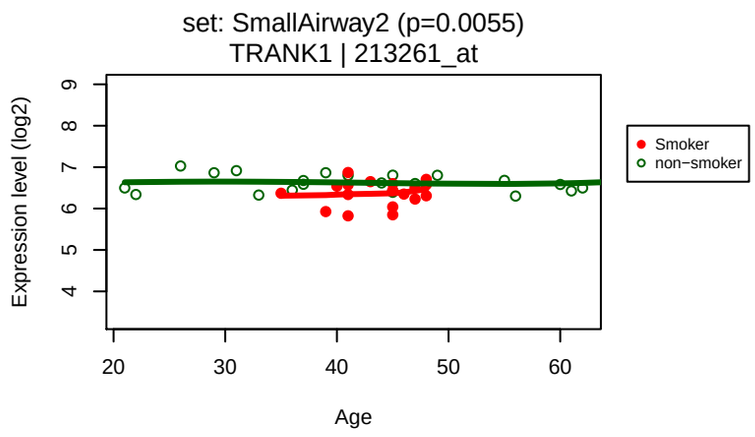
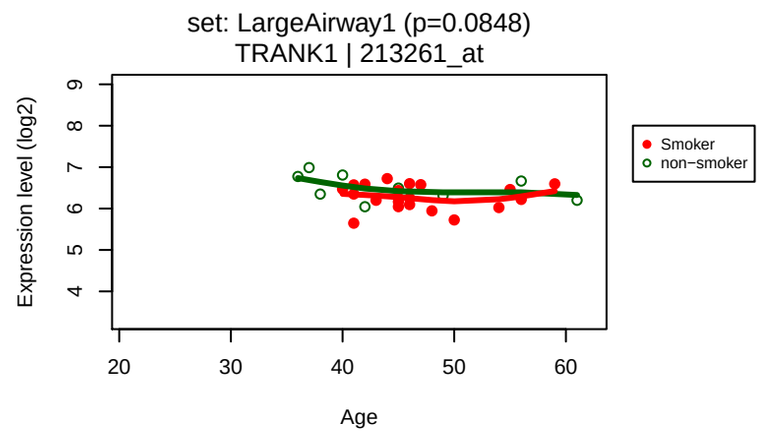

Gene expression of smokers and nonsmokers
gene: TRANK1 | 213261_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6304
p=0.0055
p=0.1497
p=0.0848
p=0.6683
p=0.6059
p=0.0021
p=0.3017
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

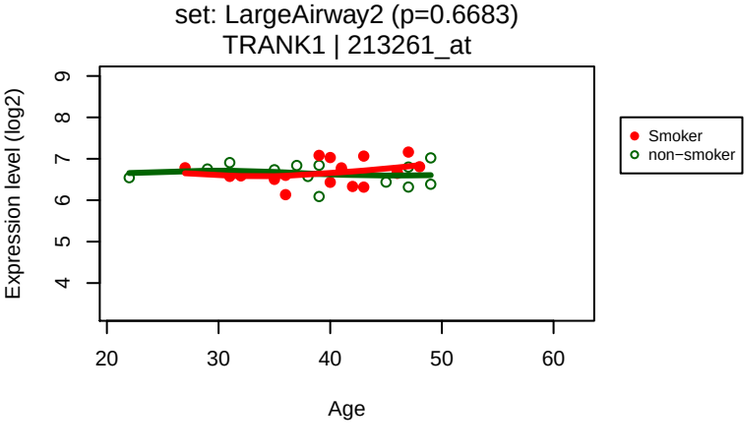

Gene expression ratio (smokers/nonsmokers)
gene: TRANK1 | 213261_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6304
p=0.0055
p=0.1497
p=0.0848
p=0.6683
p=0.6059
p=0.0021
p=0.3017
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

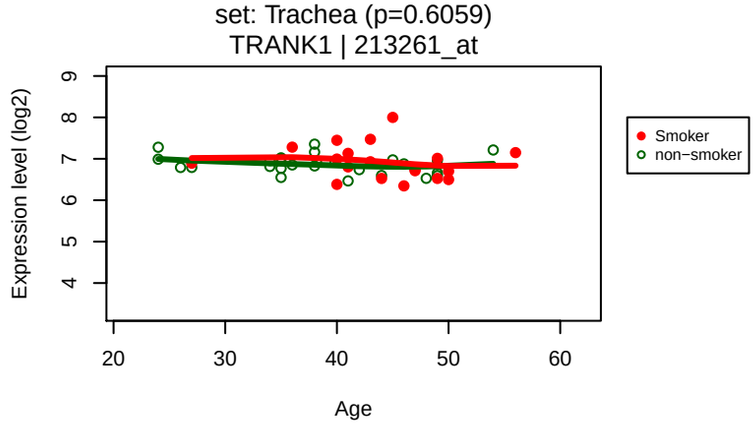

Gene expression ratio in Males
gene: TRANK1 | 213261_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0038
p=0.6223
p=0.7919
p=0.6351
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRANK1 | 213261_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8654
p=0.987
p=0.5423
p=0.3211
SmallAirway2
LargeAirway2
Trachea
Macrophage2

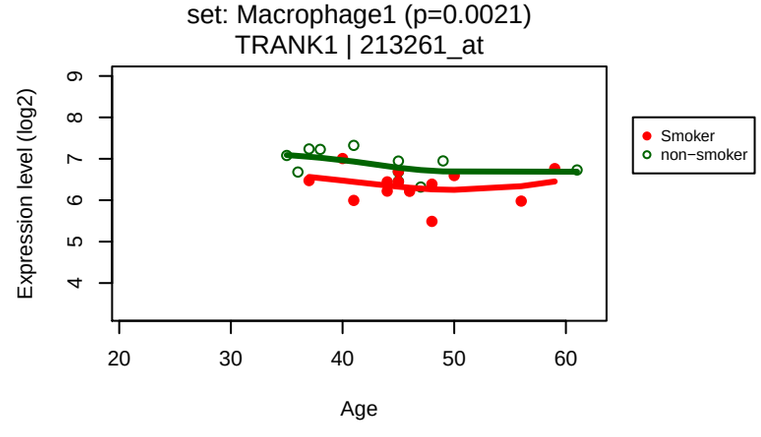

set: SmallAirway1 (p=0.6304)
TRANK1 | 213261_at
Expression level (log2)
Age
Smoker
non−smoker

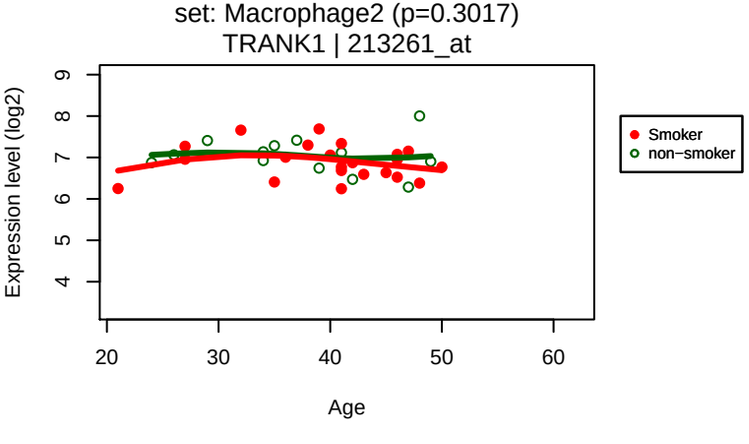

set: SmallAirway2 (p=0.0055)
TRANK1 | 213261_at
Expression level (log2)
Age
Smoker
non−smoker

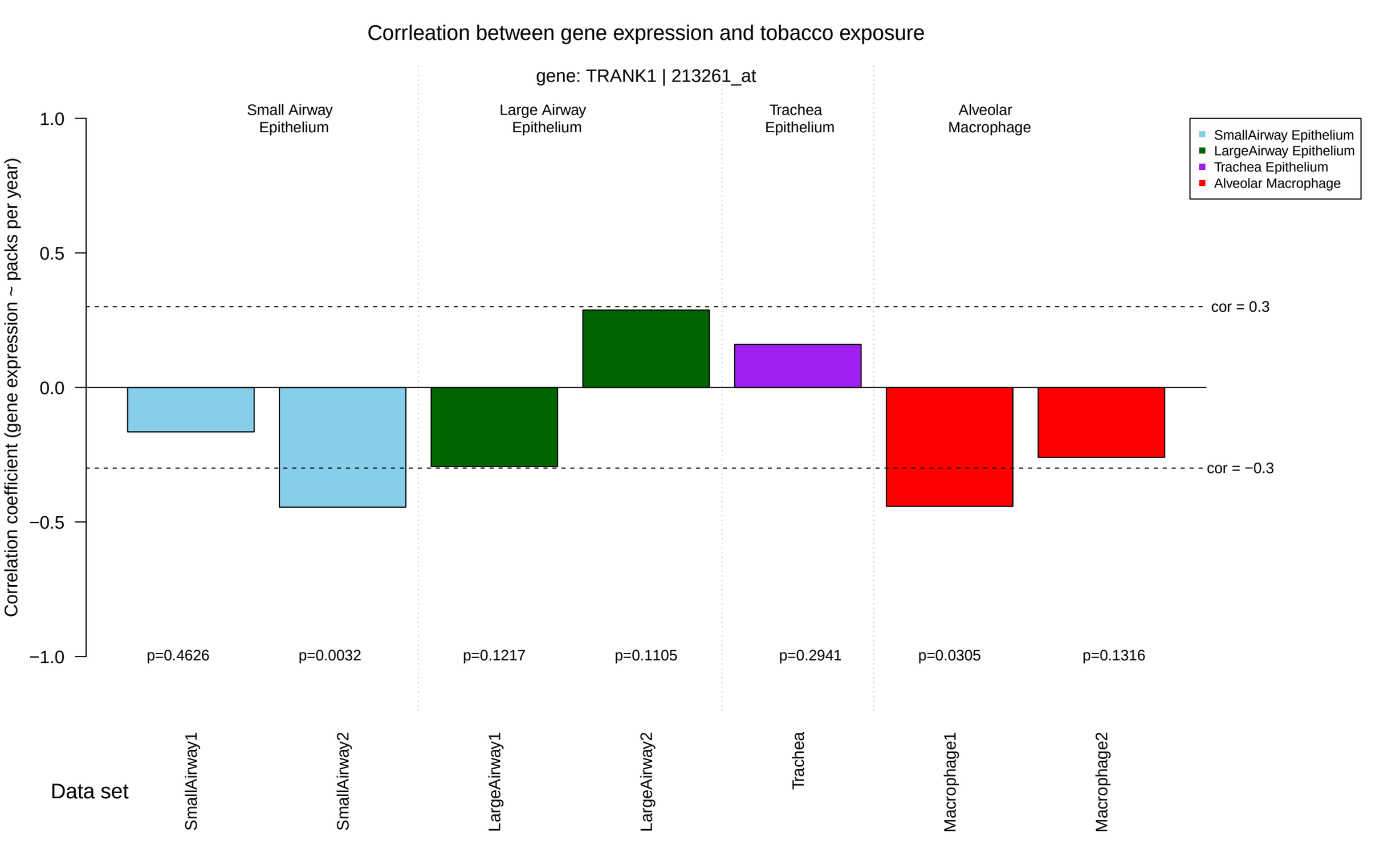

set: LargeAirway1 (p=0.0848)
TRANK1 | 213261_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6683)
TRANK1 | 213261_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6059)
TRANK1 | 213261_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0021)
TRANK1 | 213261_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3017)
TRANK1 | 213261_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRANK1 | 213261_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4626
p=0.0032
p=0.1217
p=0.1105
p=0.2941
p=0.0305
p=0.1316
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2